

MARCUS LIANG

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SKILLS SUMMARY

- **Languages:** Python, SQL, R, Java, C#, JavaScript, Node.js
- **AI & Machine Learning:** PyTorch, Scikit-Learn, LangChain, LangGraph, FAISS, OpenAI, Ollama, Prompt Engineering
- **Data Engineering & Cloud:** Databricks, Apache Spark, AWS, Snowflake, ETL Pipelines
- **MLOps & Deployment:** Docker, FastAPI, Flask, MLflow, GitHub, DataOps/DevOps,
- **Analytics, BI & Tools:** Pandas, NumPy, Power BI, Tableau, UiPath, Advanced Excel

EXPERIENCE

Business Analyst Intern, PhillipCapital, Singapore

May 2026 - Jul 2026

- Developing an AI-integrated RPA framework to not only automate standard reporting but also parse data to extract actionable business insights.
- Spearheading integration of mainstream AI tools with UiPath workflows, transforming static reports into dynamic, insight-driven analytics while saving up to 8 man-hours per reporting cycle.

Data Scientist Intern, X Star Technology Pte Ltd, Singapore

Dec 2024 - Dec 2025

- Engineered predictive repayment models deploying Random Forest and XGBoost to identify loaners' patterns and flag high-risk accounts, improving loan recovery efficiency.
- Included in developing machine learning models to analyse financial and behavioural data for credit risk assessment, increasing automated approval rates by ~80%; collaborated with cross-functional teams in Beijing.
- Architected interactive Power BI dashboards visualise repayment performance and track KPIs for dealer network, supporting decision-making.
- Executed financial Vintage Analysis grouped by Months on Books (MOB) to evaluate long-term loan performance and forecast future outcomes, such as early settlements.

Cross-Border E-Commerce Intern, Shopee Singapore Pte Ltd, Singapore

Sep 2020 - Feb 2021

- Automated key operational processes using Google Sheets and email notifications, reducing manual workload by up to 10 hours per week.
- Developed and implemented SQL queries to extract and analyse critical sales data, empowering data-driven decisions on customer behaviour and sales trends.
- Conducted in-depth analysis of monthly sales data to uncover trends and visualise insights through dynamic dashboards.

EDUCATION

National University of Singapore

Aug 2023 - Present

School of Computing, Bachelor of Science in Business Analytics (Honours)

- Specialised in Machine Learning
- Expected Graduation: Dec 2026
- Relevant Coursework: BT4221 Advanced Analytics with Big Data Technologies, BT4240 Machine Learning for Predictive Data Analytics, BT3017 Feature Engineering for Machine Learning, CS2040 Data Structures and Algorithms

Ngee Ann Polytechnic

Apr 2018 - May 2021

School of InfoComm Technology, Diploma with Merit in Financial Informatics

- Specialised in Financial Analytics
- Relevant Coursework: Deep Learning, Predictive Analytics, Applied Analytics - Finance

KEY PROJECTS

AI Engineer, Agentic Collaborative Reasoning for Cancer Research, National Cancer Centre Singapore

Jan 2026 - Apr 2026

- Architected a LangGraph multi-agent system enabling clinicians to query OMOP CDM healthcare data via natural language, running offline via Ollama to ensure patient data privacy
- Engineered a Feature Engineering Agent implementing the Refactor → Break Down → Compile decision procedure (ArXiv 2506.09742), improving Kaplan-Meier survival analysis reliability by eliminating column-name guessing
- Designed a Human-in-the-Loop (HITL) workflow with three clinician review checkpoints leveraging LangGraph interrupts, ensuring auditability before any results are delivered.
- Enhanced accuracy of clinical concept mapping by replacing traditional lexical search with a vector-based similarity search engine using ClinicalBERT and FAISS within the OMOP domain.

AI Engineer, Judicial Analytics & Legal NLP, National University of Singapore

Jan 2026 - Apr 2026

- Structured a custom 3-stage computerised data extraction pipeline utilising the OpenAI GPT-5 API to convert 122 raw, unstructured PDF court judgements into machine-learning-ready JSON records.
- Engineered an "LLM-as-a-Judge" quality assurance auditor, combining a secondary LLM with deterministic regex rules to automatically flag hallucinations and mathematically scrub target leakage.
- Parsed dense judicial narratives into a strict IRAC (Issue, Rule, Application, Conclusion) framework by employing advanced zero-shot and few-shot prompting techniques.
- Developed a custom Hierarchical Attention Network (HAN) for binary classification of corporate liability, utilising Bidirectional GRU layers and SwiGLU activation functions.